CHEMISTRY Department
SIMHADRI

Ref:SMTPP/Chemistry/2022-23/LMI CHEM/747404

Date: 09/01/2023(16:04:14)

Subject:LMI-THE CONTROL OF MARINE FOULING BY CONTINUOUS CHLORINATION

LMI/OIN/OPS/CHEM/002 " THE CONTROL OF MARINE FOULING BY CONTINUOUS CHLORINATION" was due for review. LMI has been reviewed and found **no changes.**

Revision: 06 **Revision Date:** 04.01.2021 **Review Date:** 09.01.2023

LMI word copy is enclosed .

Competent authority may pl approve the LMI for implementation.

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May please be approved.

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review the role of EMD and comment if any

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Latest OIN from CC-OS may please be attached.

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CC-OS's OIN on "The Control of Marine Fouling CORPORATE OPERATION SERVICES NTPC Limited by Continuous Chlorination " is enclosed.

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Reviewed and forwarded for approval

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OPERATION DIVISION FORMAL DOCUMENTATION SYSTEM

LOCAL MANAGEMENT INSTRUCTION-LMI/OIN/OPS/CHEM/002

Revision: 06, Revision Date: 04.01.2021

THE CONTROL OF MARINE FOULING BY CONTINUOUS CHLORINATION

Approved by: -------------------(GGM- Simhadri)
Date: ……

THE CONTROL OF MARINE FOULING BY CONTINUOUS CHLORINATION

LMI No: LMI/OIN/OPS/CHEM/002		
Revision: 06	**Revision Date:** 04.01.2021	**Review Date:** 09.01.2023

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	Dhurjati Prasad Patra	GM (O&M)
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THE CONTROL OF MARINE FOULING BY CONTINUOUS CHLORINATION

CONTENTS

1.0 INTRODUCTION

In the intake conduits of coastal power stations marine fouling may build up to a remarkable extent. Some coastal stations have had to shut down and resort to manual cleaning; in extreme cases the amount of marine fouling, consisting of barnacles, tube worms, mussels and debris, may be exceeded hundreds of tons. The greatest hazard to station operation arises from the growth of mussels as these grow to a size that they can, when detached, block or partially block the condenser tubes. This immediately causes a reduction in efficiency and may lead by fretting to perforation of the tubes. This in turn leads to the introduction of sea water into the boiler water and serious corrosion in the boilers. In order to mitigate the expected problem due to marine fouling several design aspects have been envisaged in the system. Chlorination of sea water is being done on continuous basis a) at the sea water intake well with electro-chlorinators and b) at the CW fore-bay with gas chlorinators. Continuous chlorination, daily monitoring and inspections during shut down are the main three activities which usually keeps the marine fouling at a bay.

2.0 SUPERSEDED DOCUMENTS – LMI/OIN/OPS/CHEM/002 Issue: 01 Revision:5

3.0 SCOPE

Practices are recommended for the operation of cooling water circuits to minimize fouling principally by the use of chlorination. Safe operation of Electro-chlorination plant and Gas chlorination plant are also illustrated.

4.0 BREIF THEORY OF BIO - FOULING

The cooling water of all marine and estuarine stations contains organisms which can settle and grow on cooling water (CW) circuit components and can lead to mechanical damage, corrosion, reduced flow and a reduction in heat transfer.

A) Macro fouling

Macro fouling comprises of the growth of mussels, barnacles, hydroids and other fouling organisms which usually colonize intake structures, CW intake tunnels and culverts, screen fore bays and rotary screens, condenser tube plates and occasionally the discharge tunnels. Wholesale detachment of macro organism can blanket screening plant and condenser tube plates. Uncontrolled growth of macro fouling in tunnels and culverts will rapidly reduce carrying capability and increase pumping cost. If the condenser tubes are not made from titanium, single shells which lodge in them may cause perforation due to erosion corrosion and hence, lead to raw water salt contamination of the feed water and boiler corrosion. It is unlikely that other organisms in the culverts will pose a significant problem if the preventive measures are adequate for mussels. Dislodging mussel population can be difficult and may be hazardous to plant unless the condenser

inlet is protected by shell filters which are effective in preventing detached mussel shells reaching the condenser tubes.

B) Micro fouling

Infestation of CW circuits with young mussels and other macro fouling is more common during the spring and summer, micro fouling occurs all year round. Chlorination may be required at coastal stations to control micro fouling even when there are no mussels present. Fouling of the waterside surfaces of condenser tubes by biological slimes results in impaired heat transfer with associated loss of condenser vacuum.

The process of biofilm formation involves several steps. Surfaces placed in the sea are immediately coated with a polymeric film consisting of proteins, polypeptides, polysaccharides and lipids derived from the degradation products of marine organisms. Free swimming bacteria are attracted to nutrients on the film and within 6 hours permanent attachment occurs when polymeric fibrils are exuded from bacteria within the film.

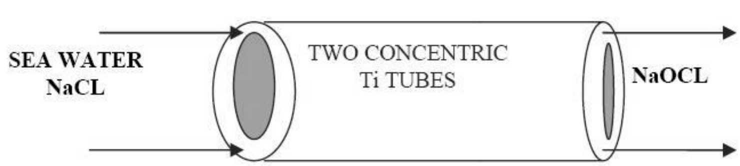
Within 7 days at temperatures of 10 to 20ºC, a layer of bacteria 50 to 100 mm or more thick will have developed on the surface. After 14 days, the slime film will have reached its maximum thickness which rarely exceeds a few hundred microns even in nutrient rich water. The biofilm so firmed can act as an anchor for any silt or mud in the water and such deposits further impair heat transfer. Titanium tubes are generally more prone to slime films than condenser tubes made from copper based alloys which are toxic to the micro organisms involved.

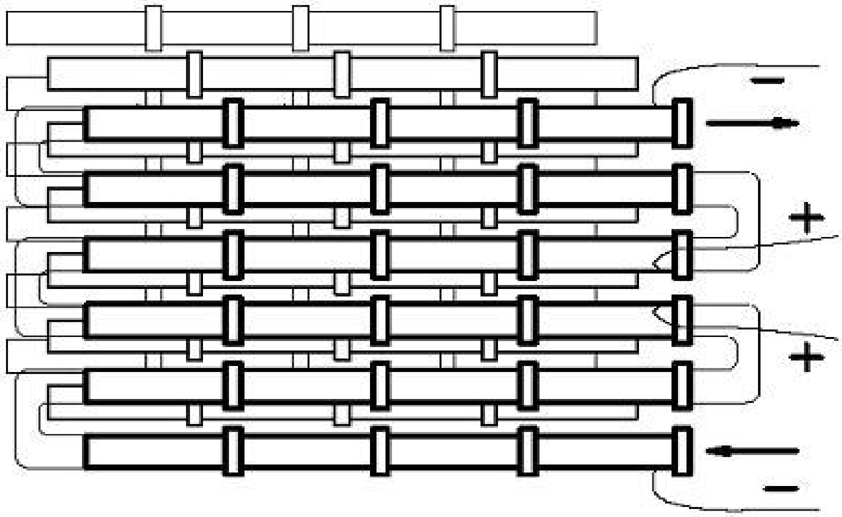
5.0 CHLORINATION TO CONTROL OF MACRO AND MICRO FOULING

5.1 ELECTROCHLORINATION SYSTEM: To control fouling due to Molluscs, algae, slime and weed at sea water intake head of inlet pipe continuous dosing of sodium Hypo chlorite to be done; otherwise they constrict the flow of Sea water in the intake pipe. Hypo chlorination also prevents sea shell deposition in the pipeline which is most troublesome to remove.

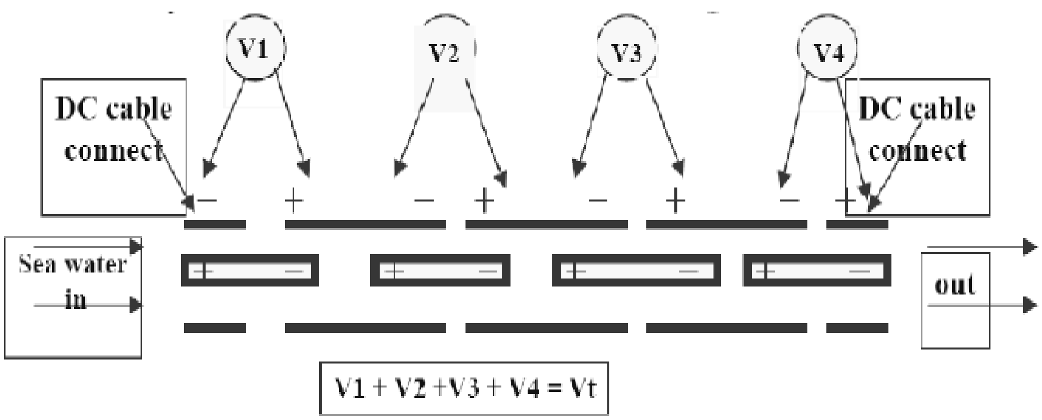
2x20 kg/hr capacity electrolyzers are available at Simhadri. There are three modules in each i.e 1A,1B,1C and 2A,2B,2C. The cells are tube shape connect electrically in series. As Sea water flows through the annular space Hypochlorite concentration increases. The concentration of Hypochlorite generated at full load is 1500ppm.

In-situ production of sodium hypochlorite in an electro chlorination plant





CHLOROPAC cells connected hydraulically in series. The module is divided into two equal sections with a series of electrical connections for individual sections. The first half section is called inlet, the second half is called the out let section. Each section contains the same number of cells. The first & the last cells in the module are connected to the -ve bus (ground potential). The two middle cells are connected to the +ve bus through the two individual poles of the DC circuit breaker. Each module requires min 5M3/hr flow to ensure turbulent flow through the cells.

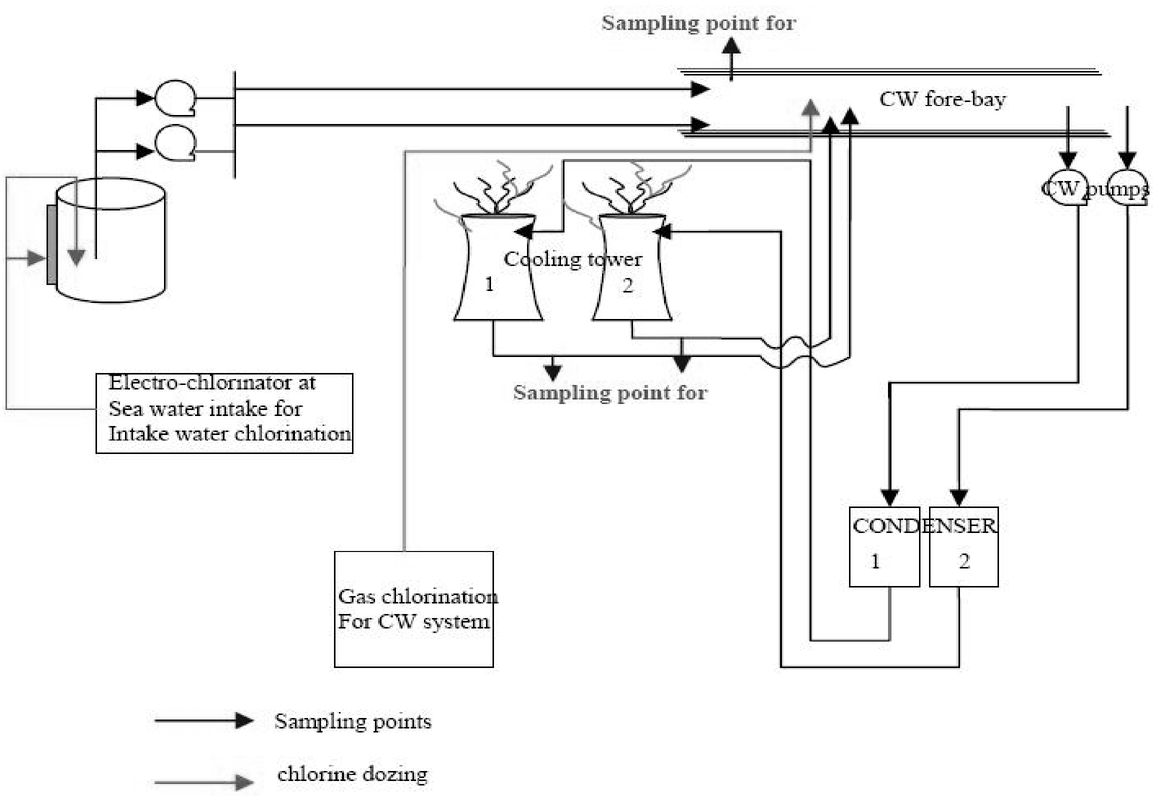


Both cathode and anode is made of Ti. The anode is coated with Platinum. The electrical cells are bipolar in design. That means the anode of every cell is directly connected to the cathodes of the next cell. The electrical connections of the electrode assembly are placed on the flanges of the electrolyser body.
Provision for Hypochlorite dosing is available at sea water pump house for both continuous and shock dosing.

5.2 CHLORINE GAS CHLORINATION:

Chlorine Gas dosing is to be done at the CW fore-bay (condenser inlet) to control bio fouling at condenser inlets, condenser outlets, cooling tower outlets to ensure that further growth organism in cooling tower and CW channel are minimized. Both continuous and shock dosing chlorination are applied at CW intake bay for effective control bio fouling. Chlorination
plant is available for both ST I & ST II CW systems.

Schematic for chlorine dozing and sampling point location (Fig – 1)



6.0 SPECIFIC OPERATING INSTRUCTIONS TO BE FOLLOWED:

6.1 ELECTRO CHLORINATION AT SEA WATER PUMP HOUSE

S. No	Instruction/ Activity	Responsibility	Records
1	One or Two electrolyzers is (are) should be kept in service round the clock so as to dose about 2 ppm of hypochlorite at the Sea water intake channel to keep the intake system i.e the lock gate, travelling water screen, well and pipe lines etc free from any marine growth.	Operation	Operator Logbook at SW P/H *(Annexure-1)*
2	The electrolyser should be controlled to produce the required level of chlorine when less than the full output is required by controlling the current from the TR unit. The current setting from the transformer rectifier controls concentration of chlorine generated.	Operation	
3	In case of non-availability of electrolyzer, dosing is to be done using sodium hypo chlorite solution stored in 10 KL tanks at sea water pump house	Operation	
4	Availability of commercial sodium hypochlorite chemical should be ensured at sea water pump house by arranging regular supply through tankers	Chemistry	
5	Dosing should be maintained in such a way that minimum of 0.5 Free residual chlorine or 600mV ORP should be maintained in sea water by the time it reaches desilting basin at CW channel inlet.	Operation	Operator Logbook at SW P/H
6	Intermittent shock dosing to be done with 5 ppm concentration once in a month or when no free residual chlorine is observed continuously for a day in sea water at desilting basin in main plant	Operation	Operator Logbook at SW P/H
7	The strainer at 800μ debris filter should be cleaned when booster pump discharge pressure falls below 7.2 KSc to avoid flow hampering to electrolyzer leading to shut down of the unit on low flow alarm.	MM-Offsite	Operator Logbook at SW P/H
8	When ECP is operational each of the three modules in an electrolyzer are sized to take sea water flow rate of 13.3 m^3/h and supplied a DC current up to 855 amps at 133VDC. It is critical to maintain the flow rate and it should be monitored to avoid overheating of modules and melting of plastic fittings and to ensure the cells remain clean and free of calcareous deposits	Operation	Operator Logbook at SW P/H
9	Healthiness and functioning of safety interlock provided for trip on low flow/ DP variation indication (if less than 4 m^3/Hr) in the	C&I,EM and Operation	Operator Logbook at SW P/H

	elctrolyser module should be checked jointly		
10	Air flow of 1000M3/Hr and pressure of 100mmWC should be maintained when one electrolyser is running to ensure that the Hydrogen liberated along with hypochlorite will be adequately diluted to less than 3%.	Operation	Operator Logbook at SW P/H
11	Functioning and healthiness of safety interlock provided to trip the entire plant and to avoid Hydrogen gas accumulation in the event of reduced air discharge should be jointly checked once in a month along with all other safety interlocks.	C&I,EM and Operation	Operator Logbook at SW P/H
12	Checking of cell wise concentration and production rate of hypochlorite at maximum DC current is to be ensured once in month	Chemistry	Water lab analysis register
13	Electrolyzers should be checked for leaks regularly and leak should be reported	Operation	Operator Logbook at SW P/H
14	Regular Monitoring DC current, water flow and temperature of electrolysers and adjustment current as per the dosing requirement should be ensured	Operation	*Operator Log sheet at SW P/H*
15	Availability of electrolysers for continuous dosing of hypochlorite should be ensured	EM,C&I, Offsite MM	
16	Reporting of deviations in maintaining FRC should be reported in daily planning meeting everyday	Operation	Operator Logbook at SW P/H

6.2 GAS CHLORINATION IN CW INTAKE BAY

S. No	Instruction/ Activity	Responsibility	Records
1	Chlorination should be done to maintain 0.5 PPM residual chlorine at outlet of condenser	Chemistry	Chlorination plant Log book (Format: Annexure-2)
2	Continuous dosing rate and shock dosing duration should be increased depending on seasonal changes in CW quality and increase in chlorine demand.	Chemistry	Chlorination plant Log book
3	Continuous chlorination should be kept in service as far as possible even when unit is not in service. Prior to stopping CW Shock chlorine dosing shall be done to get higher residual chlorine of nearly 5ppm	Chemistry	Chlorination plant Log book
4	Each stream should be run independently till the connected cylinder is completely emptied and rotation of streams should be in sequence. Always toner should be disconnected from header before lifting	Chemistry	
5	Whenever new cylinder is connected new lead gaskets should be used. After connection is done cylinder valve should be just opened and closed to detect any gas leakage using ammonia	Offsite/Chemistry	

	torch. After confirming that no gas leakages are there, Cylinder valve shall be opened half turn, and main header valve shall be opened slowly to raise header pressure.		
6	All safety appliances a) Gas canister masks, b) Breathing apparatus and c) Emergency kits should be readily available in the adjacent room. All operating and maintenance staff should be adequately trained and be very well acquainted with use of safety appliances.	Chemistry	Monthly chlorination plant checklist file
7	Mock drill for chlorine gas leakage should be conducted every year along with safety and CISF departments.	Safety	Mock drills record
8	The availability of Chlorine gas absorption system and the concentration of caustic soda in the Chlorine gas absorption system and other safety appliances should be checked once in a month and recorded	Chemistry	Monthly checklist file
9	Optimum dosing of chlorine should be done as the Chlorination discharge from the system is both a waste and a potential threat to marine ecology. In any case it is to be ensured that effluent disposed from monitoring basin should have less than 0.5 ppm residual chlorine as provided in consent order of APPCB to avoid damage to marine life	Chemistry	Analysis record of pollution lab
10	Adequate stock of liquid chlorine should be ensured through regular follow up with the supplying party/Manufacturer	Chemistry	
10	Availability of chlorination system equipment should be ensured through regular maintenance so that continuous chlorination can be done	Offsite MM/C&I/EM	
11	Reporting of deviation in maintaining residual chlorine in CW system in daily planning meeting everyday	Chemistry	

7.0 REFERENCE DOCUMENT: OIN/OPS/CHEM/002 Rev 2 Feb 2010 Reviewed on 09.03.2020

8.0 REVIEW Head of (O&M) is responsible for reviewing the document on a 2-yearly basis and as required.

Annexure-1

ELECTRO CHLORINATION SYSTEM

Time Hrs	Electrolyser no	Air Blower		Flow Through Module							Hypo Tank level cm	Dosing pump pressure Kg/Cm^2	FRC	Shock dosing	Hypo (NaOCl) Conc. ppm	External Hypo Solution tank Level Cm
		Pr mmwc	Flow M^3/Hr	Cell			AC V	AC amp	DC V	DC amp						
				1	2	3										
00:00																
04:00																
08:00																
12:00																
16:00																
20:00																

Operator's Signature				'A' Shift					'B' Shift					'C' Shift		

Annexure-2

CW CHLORINATION SYSTEM

SIMHADRI

Chlorinator No. Evaporator No. PUMP No. Date:

Time Hrs	Booster pp Discharge Pr Kg/Cm^2	Water Flow M3/Hr	Chlorine cylinder Pressure Kg/Cm^2	Evp.Temp Deg C	Chlorine Gas Pressure in Evp. Kg/Cm2	Gas Pressure before filter Kg/Cm2	Gas Pressure after filter Kg/Cm2	Gas Pressure after PRV Kg/Cm2	Gas Flow in Kg/hr	Vaccum inch in Mercury	FRC
00:00											
04:00											
08:00											
12:00											
16:00											
20:00											

Chlorine Consumption : PTW

CW

Chlorine Cylinder Stock:

Status of leak absorption System

Operator's Signature 'A' Shift 'B' Shift 'C' Shift